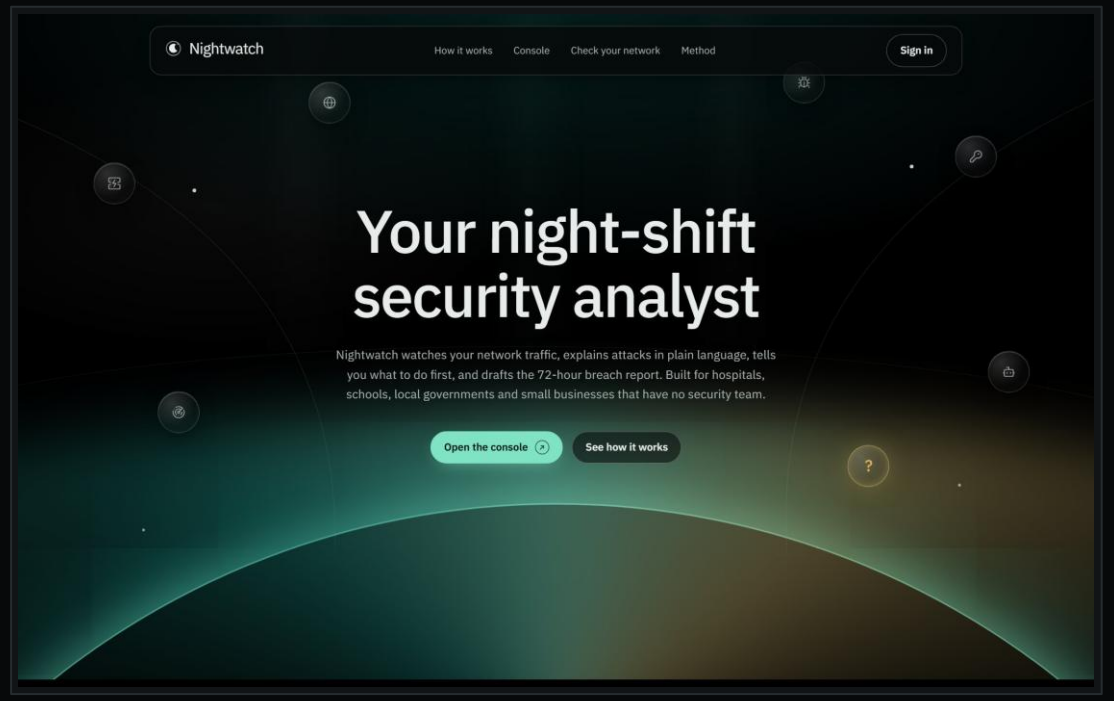
Nightwatch

Your night-shift security analyst

Network intrusion detection that explains every alert in plain language, ranks what to fix first, and drafts the breach report the law requires within 72 hours. For hospitals, schools, local governments and small businesses with no security team.



DataNova 2026 Global Data-thon
Cybersecurity and Fraud Detection, DN-CY-01

PhilHealth, 2023: the attack nobody was watching for

On 22 September 2023 the Medusa ransomware group hit the Philippine Health Insurance Corporation. About 72 workstations were encrypted, a $300,000 ransom was refused, and the stolen data was published. The antivirus licence had lapsed five months earlier.

If a national agency with an IT department missed it, a provincial hospital with one IT staff member and no night shift has little chance.

~13M

members whose data was reported compromised

72 h

to notify the National Privacy Commission after discovering a breach

46%

of security alerts are false positives, even in staffed SOCs

Sources: PhilHealth public notice UN2023-10-03; HIPAA Journal (Oct 2023); NPC Circular 16-03; Microsoft and Omdia, State of the SOC 2026. Full list in docs/RESEARCH.md.

Three jobs small organizations cannot staff

1

Notice the attack

Detection tools fire thousands of alerts a day. 73% of security teams name false positives as their top problem.

Most alerts are never looked at.

2

Decide what comes first

An IT generalist has maybe half an hour a day for security. Reviewing alerts oldest-first spends it on noise.

The real attack waits in line.

3

Report it in time

The Data Privacy Act asks for the nature of the breach, the data involved, measures taken and a contact person, within 72 hours.

Hours lost to piecing together logs.

Sources: SANS 2025 Detection and Response Survey; NPC Circular 16-03, Section 17. Full list in docs/RESEARCH.md.

Nightwatch does the analyst's night shift in five steps

1

Monitor

Reads network flow logs (CICFlowMeter format) from a live replay or an uploaded file.

2

Detect

A model trained on corrected, real attack traffic names the attack, or says it does not recognize it.

3

Prioritize

Groups alerts into incidents and ranks them Critical, High, Medium or Low.

4

Investigate

Explains each incident in plain words, with the MITRE ATT&CK technique and the evidence.

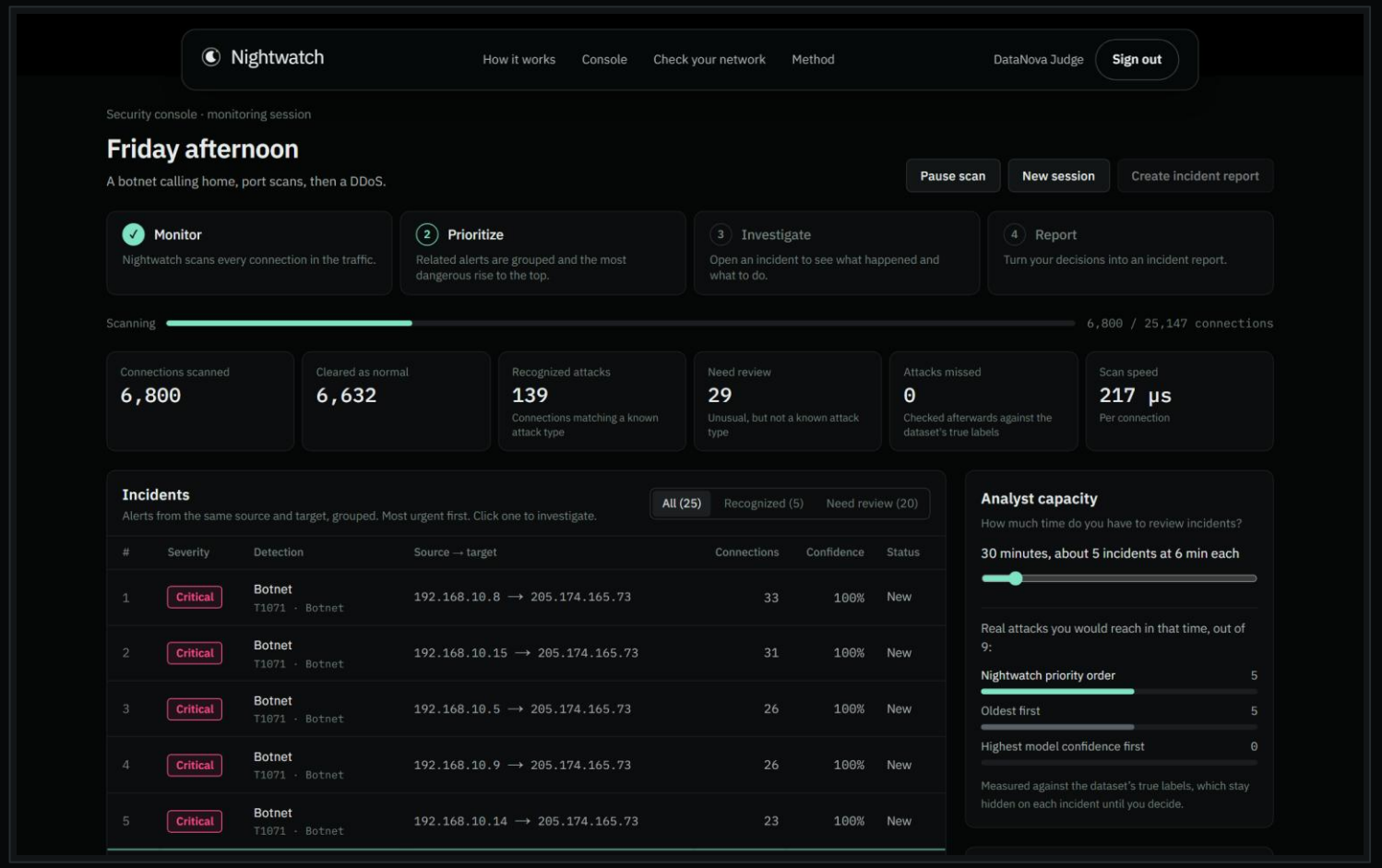
5

Report

Turns the analyst's decisions into a draft breach report and counts down its deadline.

Nightwatch never blocks traffic on its own. It tells a person what happened, how sure it is, and what to do first; the person decides.

The console: real attacks rise to the top



Live scoring

Each connection is scored as it streams in. The replay is held-out traffic the model never trained on.

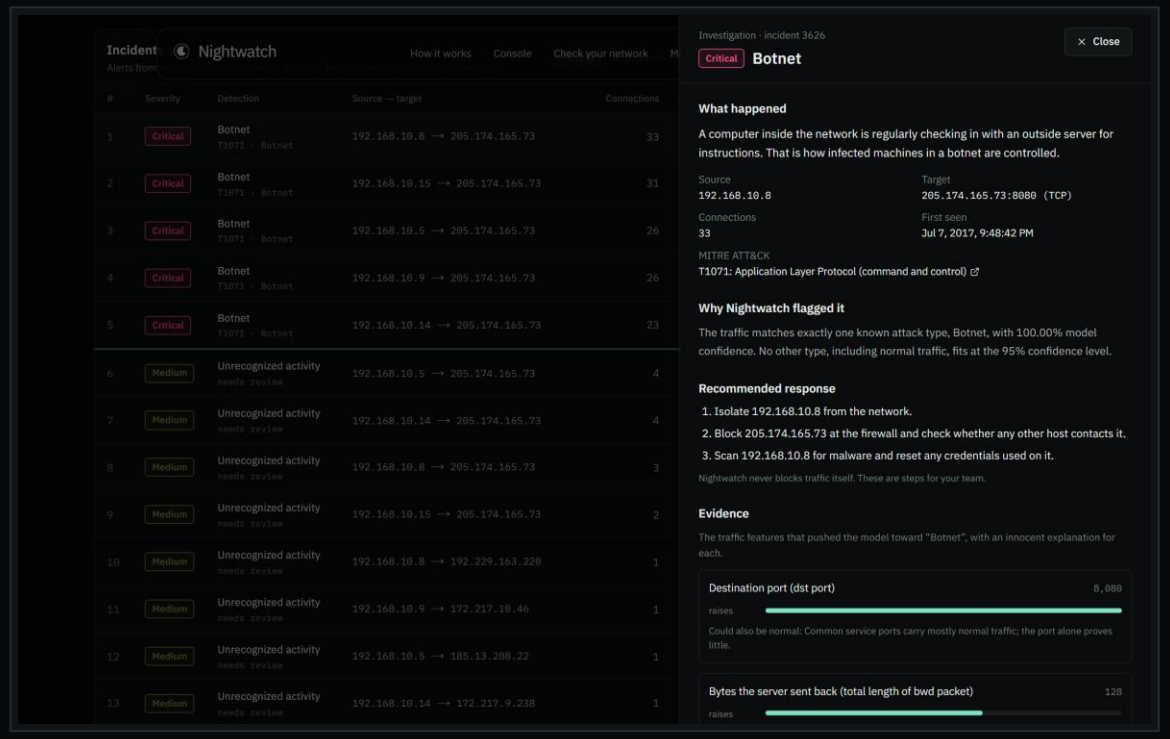
Severity first

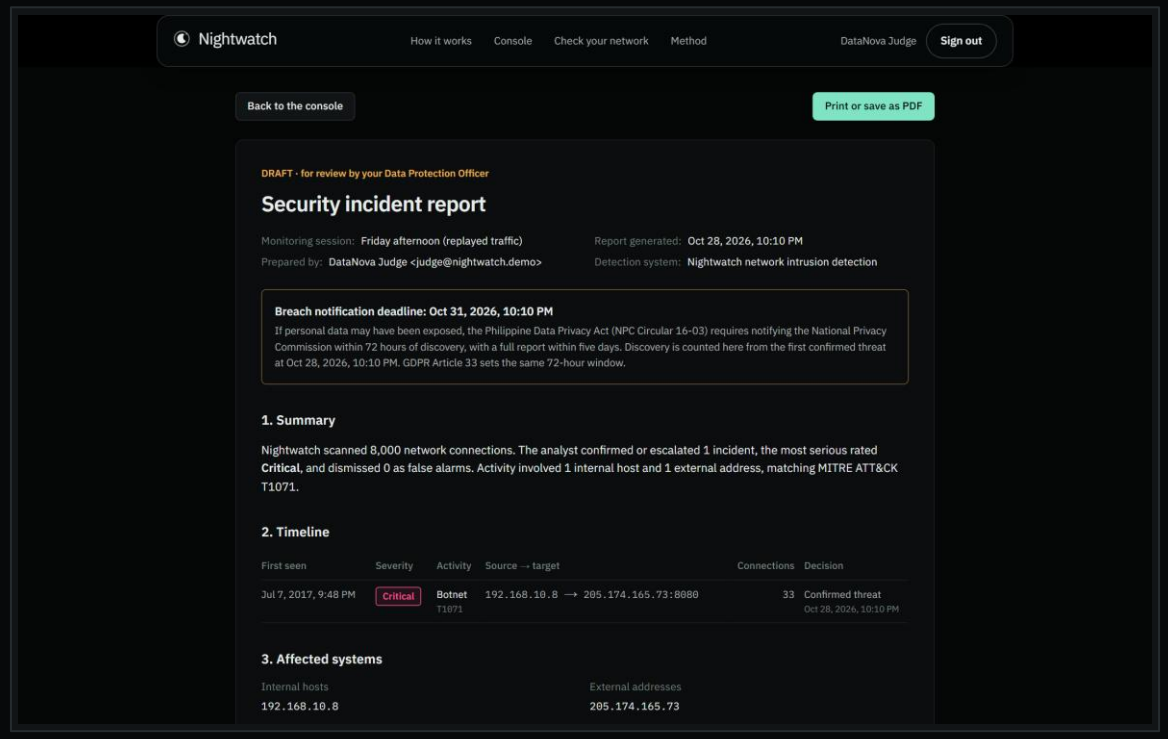
Botnet command-and-control traffic is Critical and sorts above everything else.

Your time

The capacity slider shows which incidents fit in the time you have, against oldest-first.

From one click to a breach report draft





Investigate

What happened in plain words, MITRE ATT&CK T1071, the model's evidence with an innocent explanation beside each signal, and the first response steps.

Report

Confirmed incidents become a draft with the NPC's four required fields, a timeline, affected hosts and the deadline. Marked as a draft for the Data Protection Officer.

Why 99.9% accuracy proves almost nothing

Our own model scores 99.99% accuracy on a random split. That number is mostly normal traffic being called normal. The question that matters is how well it catches each kind of attack on traffic from later in time.

>25%

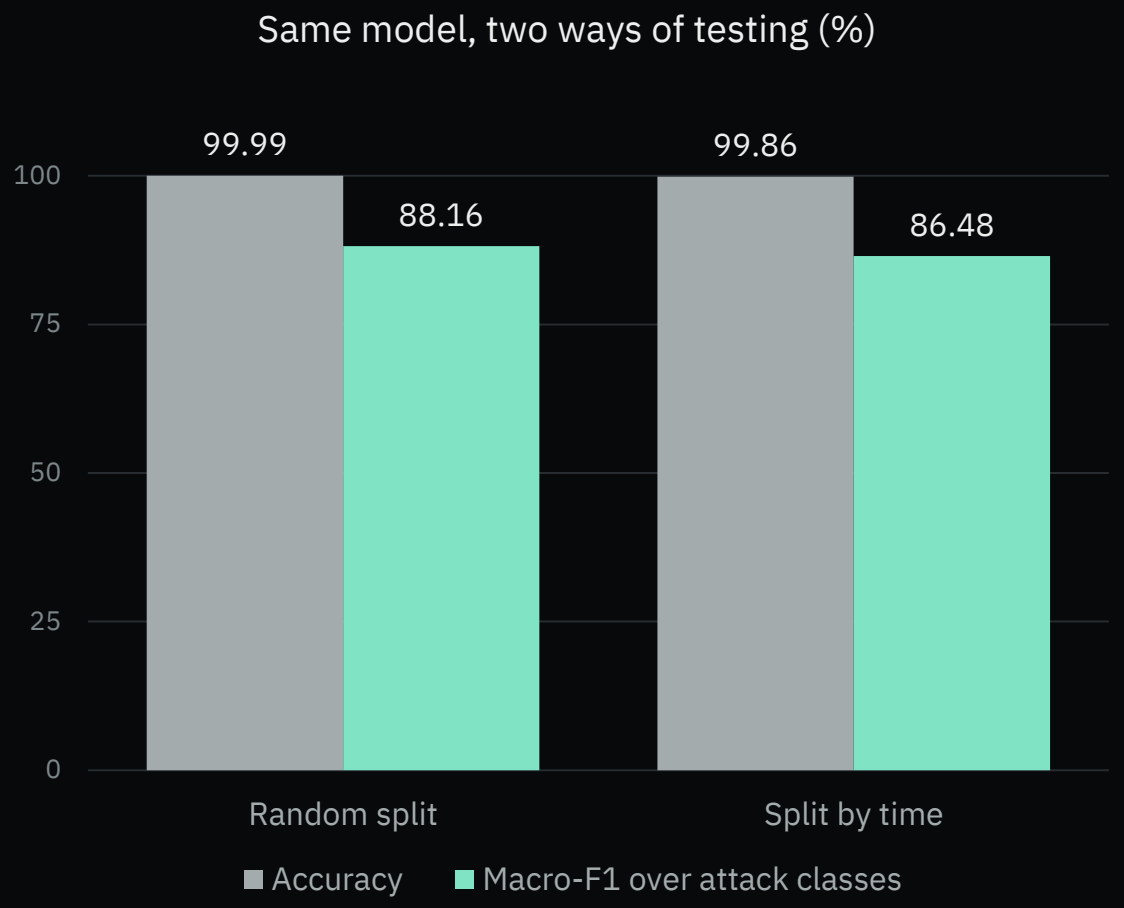
of flows in the original CIC-IDS2017 were mislabeled or broken. We use the corrected release.

30 of 30

top security ML papers had at least three evaluation pitfalls. We check ourselves against all ten.

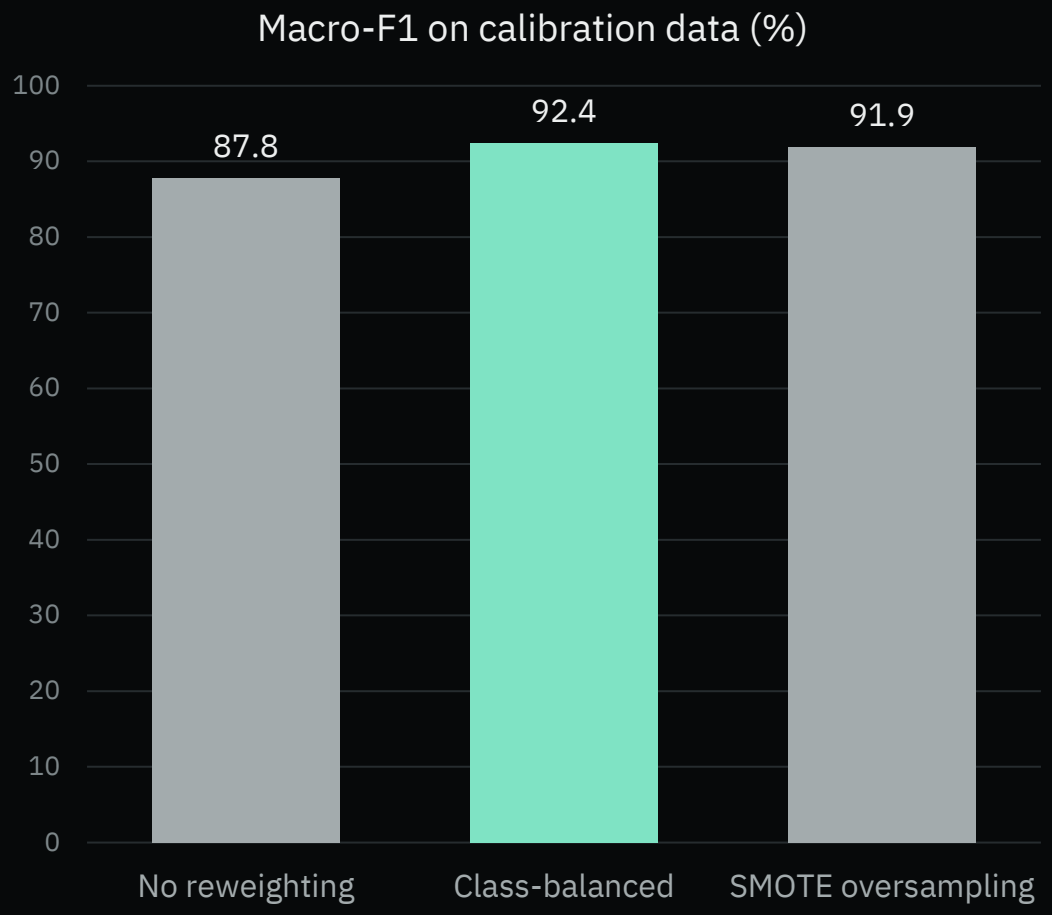
86.8%

macro-F1 with port and protocol removed. It reads behavior, not port numbers.



Sources: Engelen et al., IEEE SPW 2021; Liu et al., IEEE CNS 2022; Arp et al., USENIX Security 2022. Numbers: ml/reports/illusion.json, no_port.json.

Rare attacks get a fair vote



Rarest attacks in the test set

Attack class	Flows	Alerted	Named right
Heartbleed	2	100%	100%
Web SQL Injection	3	100%	0%
Web XSS	4	100%	75%
Infiltration	7	100%	29%
Web Brute Force	15	100%	93%
Botnet	147	100%	95%

We chose class-balanced weighting because it scored best on calibration data, before looking at the test set. Every rare attack raised an alert, but some only as Needs review: with a handful of training examples the model is not sure enough to name them.

Numbers: ml/reports/ablation.json (calibration split) and final.json (test split). Alerted: named as an attack or sent to Needs review. Named right: the single correct attack type.

It says “I'm not sure” instead of staying silent

A normal classifier must pick a label, so an attack it has never seen is quietly called normal. Nightwatch uses conformal prediction: it lists every label it cannot rule out at 95% confidence. One attack label means a recognized attack. Anything ambiguous goes to Needs review. Only flows that look clearly normal are cleared.

- **Recognized attack** — One attack type, 95% confidence
- **Needs review** — Neither a known attack nor normal
- **Cleared** — Only normal fits

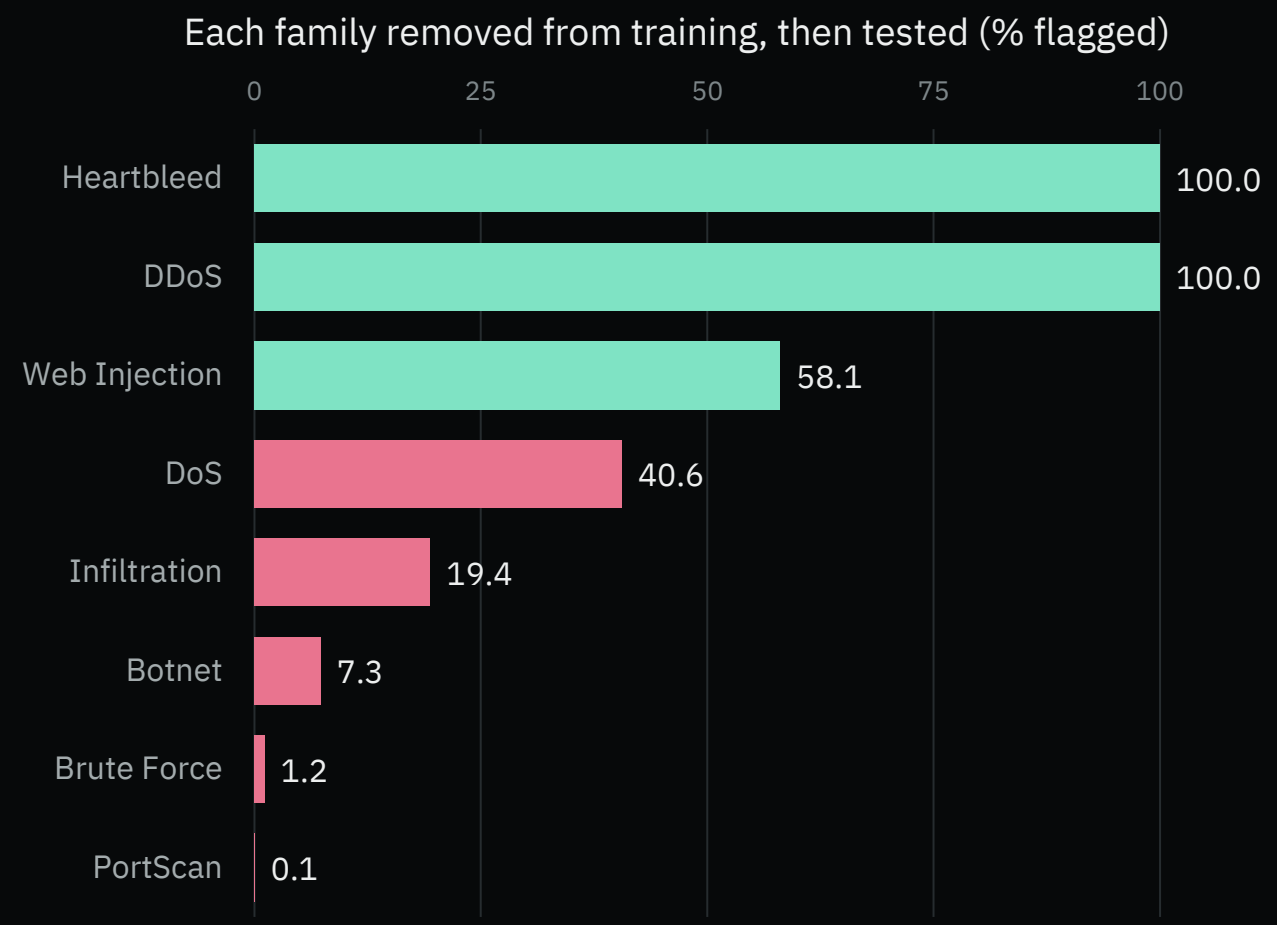
99.96% of attack flows in the test set raised an alert

0.42% of normal flows raised a false alert

0.039% of attack flows were missed

Test split: 393,993 held-out flows. Numbers: ml/reports/final.json. Method: split conformal prediction (Angelopoulos and Bates, 2021).

New attacks: what it catches and what it misses



Caught

Attacks that look structurally new: DDoS floods and the Heartbleed probe were flagged 100% of the time without ever being trained on.

Missed

Attacks that imitate normal traffic: botnet check-ins, password guessing and port scans mostly passed as normal when their family was unseen.

We show this on the site and in the FAQ. No detector catches everything; overclaiming is how organizations get surprised.

Leave-one-family-out test, model retrained eight times. Numbers: ml/reports/zero_day.json.

Thousands of alerts become a short, ranked list

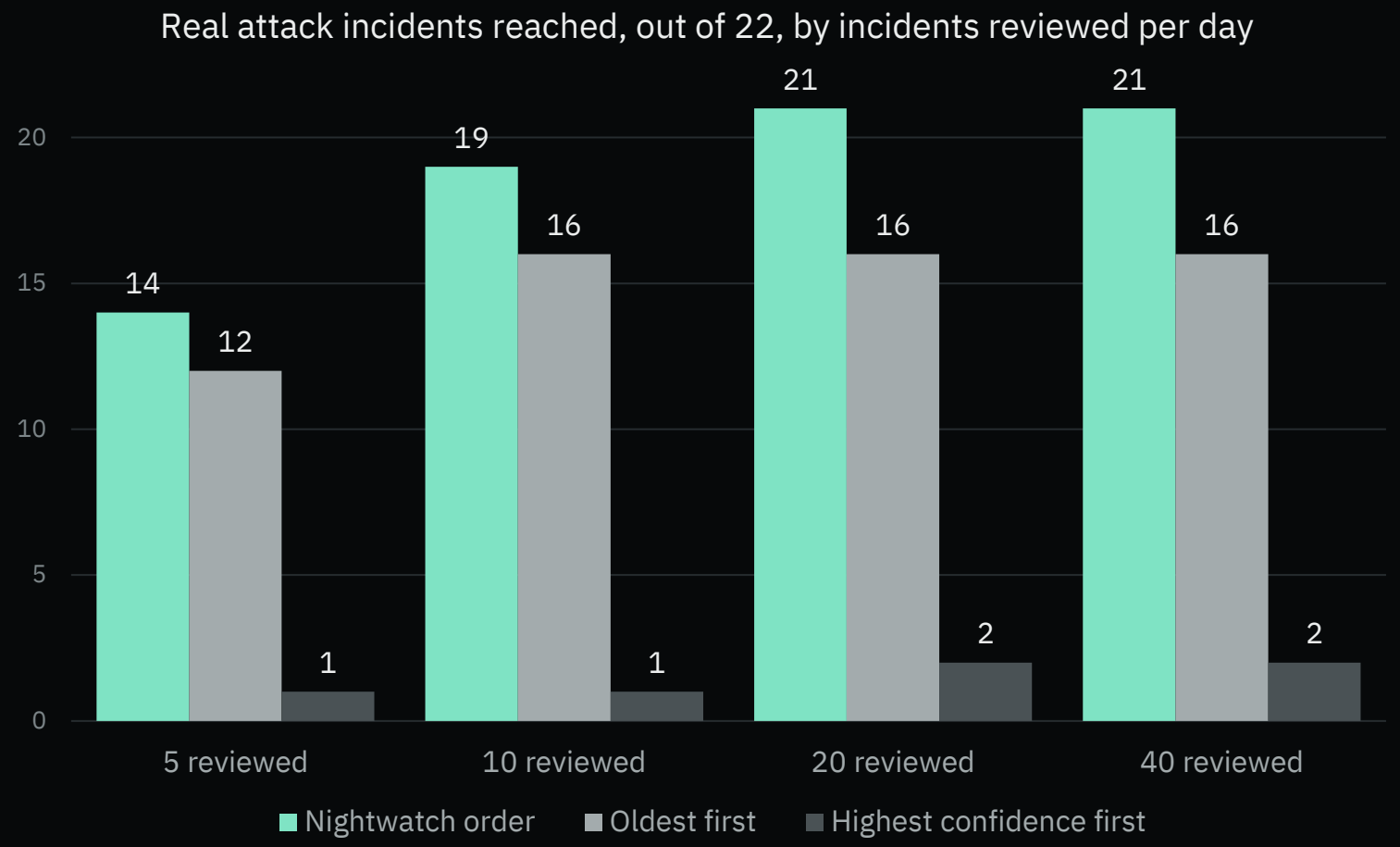
65,728

alerts on the Friday replay

126

incidents after grouping by source and target

Across all five replayed days: 102,268 alerts became 471 incidents, 217 times fewer things to look at.



Five held-out days, scored by the same ranking the console uses (severity tier first, then confidence and size). Numbers: ml/reports/triage.json.

Built to run cheaply and to be trusted

Training	Python, LightGBM, conformal calibration and an Isolation Forest, trained on the corrected CIC-IDS2017.
Scoring	The model runs in TypeScript inside the web app. It matches Python to within 0.000001 on 901 test flows.
Data and sign-in	Neon Postgres stores sessions, incidents and decisions; Neon Auth handles accounts.
Hosting	Next.js on Vercel. Free tiers cover a pilot for a small organization.

Security of Nightwatch itself

- Every data route checks the signed-in user on the server
- Uploaded logs are scored in memory and never stored
- Rate limits on uploads and session starts
- Append-only audit log of every analyst decision
- Strict security headers (HSTS, frame denial, no sniffing)
- Database credentials never reach the browser
- No automatic blocking; a person always decides

Code: ml/ (pipeline, 28 tests) and web/ (app, 12 tests including the scoring parity check).

What it takes to adopt, and what comes next

Adopt this week

- Export flow logs with the free CICFlowMeter tool
- Upload them on the Check your network page
- Review the ranked list and confirm or dismiss
- Hand the draft report to the Data Protection Officer

Known limits

- Trained on one public lab dataset from 2017
- Attacks that imitate normal traffic can slip through
- A quiet day still produced 96 low-priority items to review
- Reports are drafts, not legal advice

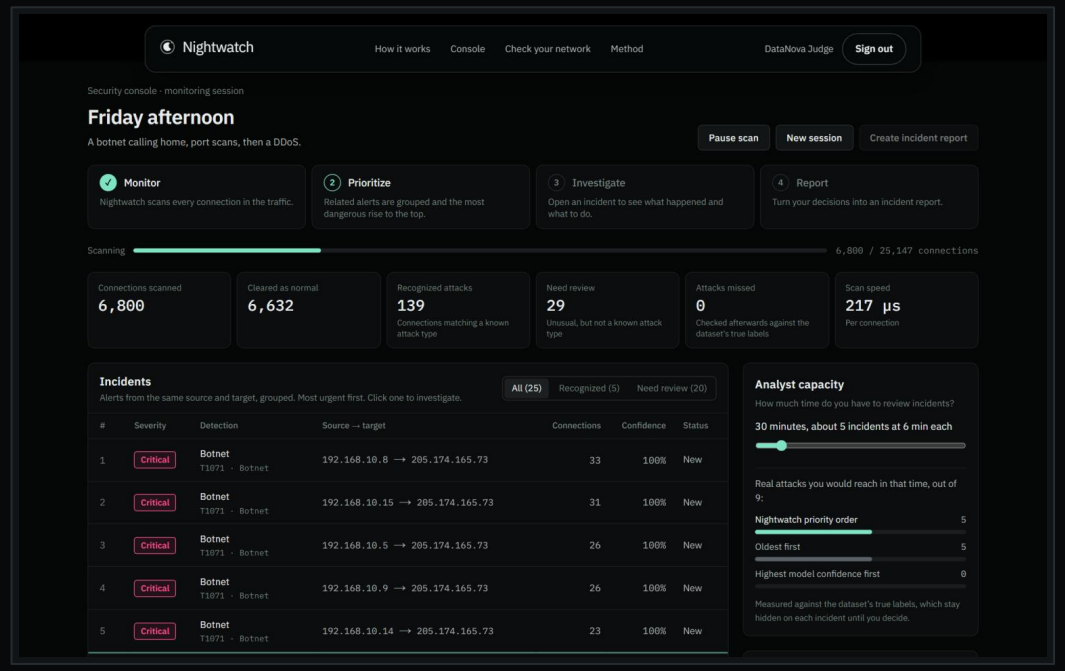
Next

- Test on the corrected 2018 dataset for drift
- Learn from analyst decisions to cut repeat false alarms
- Tune the false-alert budget per site
- Pilot with a school or local government office

Quiet-day figure: Monday replay, no attacks, ml/reports/triage.json.

Nightwatch

A security analyst for the organizations attackers count on being unwatched



Try it

nightwatch-olive.vercel.app
Press “Use the judge demo account” to sign in.

Code

github.com/yuan05-afk/nightwatch
Every number in this deck is regenerated from it.

Data: corrected CIC-IDS2017 (Liu et al., IEEE CNS 2022; Sharafaldin et al., 2018). Libraries: LightGBM, scikit-learn, imbalanced-learn, Next.js, Drizzle, Neon.